



# Majority Element 82

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## RATE THIS CHALLENGE



Given an integer array of size  $n$ , find the majority element. A majority element is an element that appears more than  $\lfloor n/2 \rfloor$  times.

If a majority element exists, print it. Otherwise, print -1. Complete the function `majorityElement` that takes the following parameter:

`int arr[n]`: an array of integers Returns `int`: the majority element if it exists; otherwise -1.

### Input Format

The first line contains an integer  $n$ , the number of elements in the array. The second line contains  $n$  space-separated integers.

### Constraints

$1 \leq n \leq 10^5$   $-10^9 \leq arr[i] \leq 10^9$

### Output Format

Print the majority element if it exists; otherwise, print -1.

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Language

C++11



```
6 using namespace std;
7 int main() {
8     int n;
9     cin >> n;
10    int arr[100000];
11    for (int i = 0; i < n; i++)
12        cin >> arr[i];
13    int candidate = 0, count = 0;
14    for (int i = 0; i < n; i++) {
15        if (count == 0)
16            candidate = arr[i];
17        if (arr[i] == candidate)
18            count++;
19        else
20            count--;
21    }
22    count = 0;
23    for (int i = 0; i < n; i++) {
24        if (arr[i] == candidate)
25            count++;
26    }
27    if (count > n / 2)
28        cout << candidate;
29    else
30        cout << -1;
31    return 0;
32 }
33
```

Line: 33 Col: 1

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